

Self-efficacy training for patients with end-stage renal disease

Shiow-Luan Tsay PhD RN

Associate Professor, Graduate Institute of Nursing, National Taipei College of Nursing, Taipei, Taiwan

Submitted for publication 17 April 2002

Accepted for publication 31 March 2003

Correspondence:

Shiow-Luan Tsay,
National Taipei College of Nursing,
365 Ming Te Road,
Pei-Tou 112 Taipei,
Taiwan.
E-mail: sltsay@ntcn.edu.tw

TSAY S.-L. (2003) *Journal of Advanced Nursing* 43(4), 370–375

Self-efficacy training for patients with end-stage renal disease

Background. Patients with end-stage renal disease often fail to follow a prescribed diet and fluid regimen, which undermines the effectiveness of care and leads to unpredictable progression of the disease and greater likelihood of complications.

Aim. The purpose of this randomized controlled study was to examine the effectiveness of self-efficacy training on fluid intake compliance.

Methods. This study took place in northern Taiwan. Eligible patients were receiving routine haemodialysis; 20–65 years of age; living in a community setting; able to read and write; and willing to participate. Sixty-two end-stage renal disease patients participated in the study. Those in the experimental group ($n = 31$) received 12 sessions of structured self-efficacy training; the control group patients ($n = 31$) received only routine care. The intervention was based on Bandura's theory and included an educational component, performance mastery, experience sharing, and stress management. The outcome measure was the mean body weight gain between dialysis sessions. Data were collected at baseline, 1, 3 and 6 months following the intervention and analysed by a descriptive and repeated-measures ANOVA.

Results. Experimental group mean weight gains decreased gradually following self-efficacy training. However, control group mean weight gains decreased only slightly over time. These results were statistically significant when baseline differences controlled for ($P < 0.05$).

Conclusions. The study supports the effectiveness of the self-efficacy training in controlling mean body weight gains of end-stage renal disease patients receiving haemodialysis.

Keywords: self-efficacy, fluid intake compliance, end-stage renal disease, weight gains, haemodialysis, nursing

Introduction

Patients with end-stage renal disease (ESRD) undergo a complex treatment regimen involving dialysis and a wide range of dietary restrictions and lifestyle changes which affect their social and psychological functioning (Tsay & Healstead 2002). Although such patients often routinely receive haemodialysis, which can remove large quantities of fluid in short periods of time, dietary and fluid intake compliance remain essential. Studies have reported that 33–50% of haemodialysis patients were non-compliant with fluid restriction

regimens (Bame *et al.* 1993, Long *et al.* 1998). This undermines the effectiveness of the treatment, resulting in unpredictable progression of the disease and greater likelihood of complications (Bame *et al.* 1993). Therefore, interventions that promote fluid intake compliance in this population are essential.

Several intervention studies have been performed to assist ESRD patients to control weight gain, including behavioural modification in children having haemodialysis (Carton & Schweitzer 1996) or adults (Hener *et al.* 1996, Laidlaw *et al.* 1999), and cognitive behavioural therapy for fluid control in

adult haemodialysis patients (Sagawa *et al.* 2001). However, in order to implement these interventions, the therapist has to be professionally trained in the field of psychology. Few nephrology nurses are qualified for such therapy and, therefore, the possibility of applying these interventions to haemodialysis patients may be limited. However, Bandura's self-efficacy theory offers an easy framework for nurses to assist and support ESRD patients in the lifestyle adjustments they need to make (Bandura 1997).

The treatment of ESRD is long-term, and patients manage their illness by engaging in self-care strategies such as dietary and fluid control. Those who have confidence in their ability to manage self-care have better fluid intake compliance (Lev & Owen 1998). Although the physiological and psychological symptoms cannot be completely relieved, interventions that are effective in increasing confidence in managing the illness are linked to increasing self-efficacy expectations. There is a growing body of literature suggesting that self-efficacy exerts a causal influence on behaviour (Bandura 1997). In Bandura's theory, self-efficacy is defined as a person's confidence in being able to perform self-care successfully to produce a desired outcome (Lev & Owen 1996, Bandura 1997). To date, no investigator has studied the relationship between a self-efficacy training programme and fluid intake compliance in managing chronic illness. However, correlation studies suggest that increased self-efficacy is associated with adherence to treatment, health-promoting behaviour, and decreased physical and psychological symptoms (Cummings *et al.* 1982, Given & Given 1989, Lev 1997, Tsay & Healstead 2002). Inability to adjust to illness may lead to negative outcomes such as treatment non-compliance.

Fluid intake compliance is usually measured by mean weight gain between dialysis sessions (Rosenbaum & Smire 1986). Accurate measurement of the body weight of patients with ESRD is important because it has been shown to correlate with the adequacy of dialysis delivery and disease prognosis (Cooper *et al.* 2000). Recent studies have documented that bioelectrical impedance analysis (BIA) and maximal inferior vena cava (IVC) diameter in ultrasonography correlate well with the dry weight of ESRD patients (Zhu *et al.* 1999, Cooper *et al.* 2000, Yanagiba *et al.* 2001). However, these new methods need to be further investigated before applying them to the ESRD population.

Patients who suffer from ESRD must often be on dialysis for 3–5 hours and for 3 days a week. However, many fail to follow the strict diet and fluid restrictions required, for example, type and quantity of food and salt and fluid intakes. Researchers have found that one-third of ESRD patients are non-compliant with fluid restrictions (Schneider *et al.* 1991,

Long *et al.* 1998). In a larger study ($n = 1250$, 29 centres), Bame *et al.* (1993) documented that 49.5% of the haemodialysis patients were non-compliant.

Factors associated with compliance have been reported. For example, Rosenbaum and Smire (1986) found that perceived self-efficacy to resist the urge to drink had a significant positive relationship with adherence to fluid restrictions imposed during haemodialysis. Patients who were convinced that they would be able to resist the urge to drink, despite their thirst, were more likely to adhere to fluid restrictions. Brady *et al.* (1998) found that those with higher self-efficacy had lower mean weekend interdialysis fluid weight gains.

In summary, many ESRD patients have difficulty complying with fluid restrictions. These patients require major lifestyle changes and self-efficacy to comply with the treatment regimen of their chronic illness. A person with an increased perception of self-efficacy is more likely to participate in self-care activities and thus increase their adherence to the treatment regimen. Although self-efficacy accounts for outcomes in illness, little research has addressed methods specifically aimed at enhancing this in patients having haemodialysis.

The study

Aims

This study examined the effects of a self-efficacy training programme on fluid intake compliance as measured by mean weight gain in chronic dialysis patients. The study provides a theory based interventional model for nurses who care for non-compliant dialysis patients.

Design

An experimental design was used to investigate the effectiveness of self-efficacy training on fluid intake compliance among ESRD patients receiving routine haemodialysis.

Sample

The population of interest was ESRD patients. Patients were recruited from three outpatient haemodialysis centres at major hospitals in northern Taiwan. Eligible patients were receiving routine haemodialysis three times a week; able to walk and eat without assistance; at least 18 years of age; living in a home setting; and were willing to participate. Hospitalized individuals and those with acute illness, psychological or cognitive disorders, or physical limitations in

self-care were excluded. The sample size needed was calculated using tabulation scheme for sample estimation with repeated measures design, as recommended by Barcikowski and Robey (1984) and Stevens (2002). According to the table (Stevens 2002, p. 511), for a power of 0.80, average correlation of 0.50, a medium effect size of 0.35, four repeated measures and α of 0.05, 27 subjects were needed for each group. Allowing for an attrition rate of 20%, an estimated 32 subjects per group were needed, giving a total sample of 64.

Method

A total of 80 subjects were approached and 64 agreed to participate, giving a response rate of 80%. Once participants who agreed to enrol in the study had signed a consent form, they were asked to complete the demographical and medical data form, and then blind randomization to either a self-efficacy training programme plus usual care ($n = 32$) or to routine care ($n = 32$) was carried out. Only the researcher knew which treatment patients were receiving, and care providers (physicians, nurses, dieticians, social workers) were not informed of participants' treatment groups. One patient from each group dropped out because of hospitalization or relocation during the study. Thus, a total of 62 finished the study.

Patients received their dialysis on a first come, first served basis depending on the availability of beds at that time. Those in the experimental group were asked not to discuss the content of the self-efficacy training with other patients. Thus, the treatment diffusion effects between the groups were limited.

Mean weight gains at baseline, 1, 3 and 6 months following the intervention were calculated. Weight gain outcomes were the means of four measurements for all patients at 1 month, eight measurements for all patients at three months, and 12 measurements for all patients at 6 months. Participants received three haemodialysis treatments each week throughout the study period from August 2000 to July 2001 and were weighed using a calibrated scale before and after each dialysis session.

Self-efficacy intervention

A structured, individualized training programme created by the investigator was implemented for the treatment group. The aim was to improve patients' achievements and experiences, and to use verbal persuasion through praise and encouragement, and decrease emotional and physical arousal through the intervention. The programme consisted of

12 sessions, each lasting 1 hour, and conducted three times per week by two trained nephrology nurse specialists while patients were receiving dialysis. The programme focused on topics such as the pathophysiology of renal failure and haemodialysis, medications, complications, nutrition, fluid restriction, control of thirst and urge to drink, and stress management. Participants also learned to relax muscles through listening to audiotaped instructions. In addition, they were interviewed about their dietary habits and fluid intake, and the factors associated with weight gain were assessed and discussed. Participants were encouraged to set attainable objectives such as 'decreasing a cup of tea or water a day'. If the goals were achieved, praise and recognition rewards were given. Individual counselling sessions were offered stressing physical and emotional adjustment to the chronic illness. Participants recorded their food and liquid intake daily, and these records were reviewed during each treatment.

Outcome measures

Fluid intake compliance was measured by mean weight gains between dialysis sessions. Postdialysis weight was subtracted from the weight preceding the next dialysis session, indicating the weight gain between sessions. This is considered to be the most reliable and valid measure of fluid intake compliance among dialysis patients (Rosenbaum & Smire 1986).

Ethical considerations

Following ethics review board approval, the researcher and an assistant explained the study to potential participants, including its purpose, methods, benefits and risks. Anonymity and confidentiality were assured and participants were told that they could withdraw from the study at any point without implications for their subsequent treatment.

Data analysis

Data were analysed using the SPSS 10.0 version of Windows software. Descriptive statistics were generated for sample characteristics and weight change variables. A repeated-measures ANOVA was used to test the effectiveness of the intervention on mean weight gains between groups over time.

Findings

Analysis of the sample

Table 1 lists the clinical and demographical characteristics of patients who completed the study ($n = 62$). The typical

Table 1 Demographical and clinical characteristics ($n = 62$)

Variable	Self-efficacy group ($n = 31$)	Control group ($n = 31$)
Age (years)	57.51 $\pm$ 11.41	57.94 $\pm$ 11.62
Time on dialysis (months)	35.19 $\pm$ 32.82	41.64 $\pm$ 40.04
Number of comorbid conditions	0.99 $\pm$ 0.96	0.92 $\pm$ 0.77
Kt/V	1.41 $\pm$.22	1.40 $\pm$ 0.23
Serum albumin (mg/dL)	3.73 $\pm$ 0.51	3.88 $\pm$ 0.81
Haematocrit (%)	9.77 $\pm$ 1.38	9.55 $\pm$ 1.40
Haemoglobin (mg/dL)	29.96 $\pm$ 4.34	29.17 $\pm$ 4.28

participant had a mean age of 57.77 years ($SD = 11.23$), and was female (58.1%), married (96.8%), and retired or unemployed (87.1%). Many had graduated from elementary school (45.2%) and had a religious faith (69.4%). Mean renal disease severity was moderately severe (mean = 6.62, $SD = 4.42$, range = 1–10), and mean length of dialysis was 40.79 months ($SD = 34.43$). Many participants complained of sleep disturbances (71%), fatigue (69.4%), pruritus (67.7%), muscle cramps (56.5%) and joint pain (51.6%). There were no statistically significant differences in gender, age, education levels, current use of medication, length of dialysis, symptoms, biochemical data, urea kinetic modelling (Kt/V), types of dialyser used and number of chronic diseases between the groups ($P > 0.05$). However, baseline body weight change was significantly different between the groups ($t = 4.03$, $P = 0.01$). Therefore, baseline differences in weight gain between groups were taken into account in the repeated-measured analysis by using the baseline values as a covariate, rather than as the first time point measure.

Mean weight change between groups

Descriptive statistics of mean body weight gains for the experimental and control groups at baseline, 1, 3 and 6 months following the self-efficacy training are presented in Table 2. Mean weight gains in the experimental group

Table 2 Mean and SD of patient's mean weight gains over time ($n = 62$)

Weight gains in kg	Self-efficacy group ($n = 31$)		Control group ($n = 31$)	
	Mean	SD	Mean	SD
Baseline	3.30	0.71	2.64	0.74
1 Month	3.03	0.84	2.67	0.90
3 Months	2.70	0.83	2.57	0.97
6 Months	2.58	0.71	2.58	0.95

decreased by 0.27 kg in the first month, and further reduced by 0.33 kg in the third month. These weights continued to drop by 0.12 kg during the sixth months following the self-efficacy training. However, the control group's mean weight gain decreased only slightly by 0.08 kg for the first month and further decreased by 0.11 kg at the third month; however, it increased by 0.02 kg at the sixth month.

A repeated-measures ANOVA was conducted to test for the effectiveness of the self-efficacy training. The baseline mean weight gains were applied as a covariate in the analysis. Main and group $\times$ time interaction effects were tested using the multivariate criterion of Wilk's Lambda (Λ). The results were not statistically significant (Wilks' $\Lambda = 0.98$, $F = 0.364$, $P > 0.05$; Wilks' $\Lambda = 0.97$, $F = 0.67$, $P > 0.05$), indicating that the effect of the intervention was not related to time or maturation factor.

The group main effect was significant between groups, $F(1, 56) = 8.10$, $P = 0.006$. This result supported the hypothesis that ESRD patients who received the self-efficacy training would have better fluid intake compliance than those who did not receive the training.

Discussion

The hypothesis that the experimental group, which received self-efficacy training, would have better fluid compliance than the control group was supported. The repeated-measures ANOVA showed that the experimental group patients significantly decreased their weight gains and continued to do so up to 6-months following the self-efficacy intervention. However, while we suggest that the introduction of the self-efficacy programme was reasonably successful, it is difficult to identify its actual effectiveness because the clinical significance of the improvement is unknown. Nevertheless, our results suggest that self-efficacy training might have an important role in managing patients with ESRD who are receiving haemodialysis.

Few research studies have evaluated self-efficacy training in dialysis patient outcomes; thus, the possibility of a comparison of our results with existing literature is limited. The self-care self-efficacy training was designed according to Bandura's Self-Efficacy Theory, and our significant findings support the theory that patients who receive self-efficacy training are more confident in their ability to engage in health promotion behaviours, and more compliant with fluid intake restrictions.

The sample was drawn from the northern part of Taiwan. Therefore, generalizability of the results to other samples of dialysis patients from other geographical areas cannot be guaranteed. We only monitored weight gains for 6 months

What is already known about this topic

- Patients with end-stage renal disease often routinely receive haemodialysis, which can remove large quantities of fluid in short periods of time; therefore, dietary and fluid intake compliance remain essential.
- Patients with poor compliance undermine the effectiveness of the care, resulting in an unpredictable progression of the disease and a greater likelihood of complications.
- Studies have reported that 33–50% of haemodialysis patients were in non-compliance with fluid restriction regime.
- Interventions such as cognitive behavioural therapy have attempted to promote fluid intake compliance in this population. However, few nephrology nurses are qualified for such therapy.

What this study adds

- This study examined the effects of a self-efficacy training programme, which can be easily applied by nurses to promote fluid intake compliance to dialysis patients.
- The results of this study revealed that patients in the self-efficacy training group significantly decreased their mean weight gains more than the control group patients for up to 6 months following the intervention.
- The study provides a theory-based interventional model for nurses who care for non-compliant dialysis patients.

following the self-efficacy training, and other researchers should carefully consider the duration of their studies. It is important to monitor weight gain for long periods of time to clarify how lasting the effects of self-efficacy training might be for haemodialysis patients.

Conclusions and implications for practice

Our results on self-efficacy might have important implications for ESRD patients who are on dialysis treatment. They suggest that assessment of haemodialysis patients' self-efficacy and fluid intake compliance should be an essential part of nursing practice. Clinicians should consider providing self-efficacy training to improve haemodialysis patients' confidence in controlling their weight gain between dialysis sessions.

Nurses are in a good position to influence the self-efficacy of dialysis patients and can encourage them to adopt self-care strategies by constructing situations in which they can

achieve their goals in a supportive environment (performance mastery). Nurses can assist patients by setting and supporting realistic daily performance goals (verbal persuasion), and providing appropriate anticipatory guidance (sharing vicarious experiences). They can assess progress, clarify symptom management, and respond to specific questions during consultation sessions. They can also influence patients' self-efficacy and performance by teaching them stress reduction methods to control emotional and physical arousal. The self-efficacy training programme should include coping training to improve confidence in carrying out self-care behaviours such as fluid control.

This study provides a foundation for conducting future studies of self-efficacy training for managing chronic dialysis, but replication and expanded studies are needed to allow us to say with more confidence that the self-efficacy training programme was solely responsible for improving fluid compliance. What can be said now, however, is that the fluid intake compliance programme in this study, designed using the theory of self-efficacy, was effective in decreasing weight gain up to 6 months following intervention.

Acknowledgement

The National Science Council of Taiwan provided funding for this study (NSC 89-2314-B-227-006).

References

- Bame S.I., Peterson N. & Wray N.P. (1993) Variation in hemodialysis patient compliance according to demographic characteristics. *Social Science and Medicine* 37, 1035–1043.
- Bandura A. (1997) *Self-Efficacy: The Exercise of Control*. Freeman, New York.
- Barcikowski R.S. & Robey R.R. (1984) Decisions in a single group repeated measures analysis: statistical tests and three computer packages. *American Statistician* 38, 248–250.
- Brady B.A., Tucker C.M., Alfino D.G., Tarrant D.G. & Finlayson G.C. (1998) An investigation of factors associated with fluid adherence among hemodialysis patients: a self-efficacy theory based approach. *Annals of Behavioral Medicine* 19, 339–343.
- Carton J.S., & Schweitzer J.B. (1996) Use of a token economy to increase compliance during hemodialysis. *Journal of Applied Behavior Analysis* 29, 111–113.
- Cooper B.A., Aslani A., Ryan M., Zhu F.Y.P. & Ibels I.S. (2000) Comparing different methods of assessing body composition in end-stage renal failure. *Kidney International* 58, 408–416.
- Cummings K., Becker M. & Kirscht J. (1982) Psychosocial factors affecting adherence to medical regimes in a group of hemodialysis patients. *Medical Care* 20, 567–580.
- Given B.A. & Given C.W. (1989) Compliance among patients with cancer. *Oncology Nursing Forum* 16, 97–103.

- Hener T., Weisenberger M. & Har-Even D. (1996) Supportive versus cognitive-behavioral intervention programs in achieving adjustment to home peritoneal kidney dialysis. *Journal of Clinical and Consulting Psychology* **64**, 731–741.
- Laidlaw J.K., Beeken J.E., Whitney F.W. & Reyes A.A. (1999) Contracting with outpatient hemodialysis patients to improve adherence to treatment. *ANNA Journal* **26**, 37–40.
- Lev E.L. (1997) Bandura's theory of self-efficacy: application in oncology. *Scholarly Inquiry for Nursing Practice* **11**, 21–36.
- Lev L. & Owen S.V. (1996) A measurement of self-care self-efficacy. *Research in Nursing and Health* **19**, 421–429.
- Lev E.L. & Owen, S.V. (1998) A prospective study of adjustment to hemodialysis. *ANNA Journal* **25**, 495–504.
- Long J.M., Kee C.C., Graham M.V., Saethang T.B. & Dames F.D. (1998) Medication compliance and the older hemodialysis patient. *ANNA Journal* **25**, 43–49.
- Rosenbaum M. & Smire K. (1986) Cognitive and personality factors in the delay of gratification of hemodialysis patients. *Journal of Personality and Social Psychology* **51**, 357–364.
- Sagawa M., Oka M., Chaboyer W., Satoh W. & Yamaguchi M. (2001) Cognitive behavioral therapy for fluid control in hemodialysis patients. *Nephrology Nursing Journal* **28**, 37–39.
- Schneider M.S., Friend R., Whitaker P. & Wadhwa N.K. (1991) Fluid noncompliance and symptomatology in end-stage renal disease: cognitive and emotional variables. *Health Psychology* **10**, 209–215.
- Stevens J.P. (2002) *Applied multivariate statistics for the social sciences*. Lawrence Erlbaum Associates, Hillsdale, NJ.
- Tsay S.L. & Healstead M. (2002) Self-efficacy, depression, and quality of life among patients receiving hemodialysis in Taiwan. *International Journal of Nursing Studies* **39**, 245–251.
- Yanagiba, S., Ando, Y., Kusano, E. & Asano, Y. (2001) Utility of the inferior vena cava diameter as a marker of dry weight in non-oliguric hemodialyzed patients. *ASAIO Journal* **47**, 528–532.
- Zhu, F., Schneditz, D. & Levin N.W. (1999) Sum of segmental bioimpedance analysis during ultrafiltration and hemodialysis reduces sensitivity to changes in body position. *Kidney International* **56**, 692–699.